Carbon-based nanomaterials and instruments

5. Bee Chems: Nano-silica products

6. Bilcare: Unique security technology called non-Clonable using composites to provide a fool-proof security system

7. Bottom Up Technology Corp.: Carbon nanotubes

8. Dabur Pharma: Drug delivery

9. Egoma Technologies: Nanopowders

10. Eris Technologies: Nano-education and certification

11. Icon: Analytical instruments, with a focus on nanotechnology and related analytical techniques

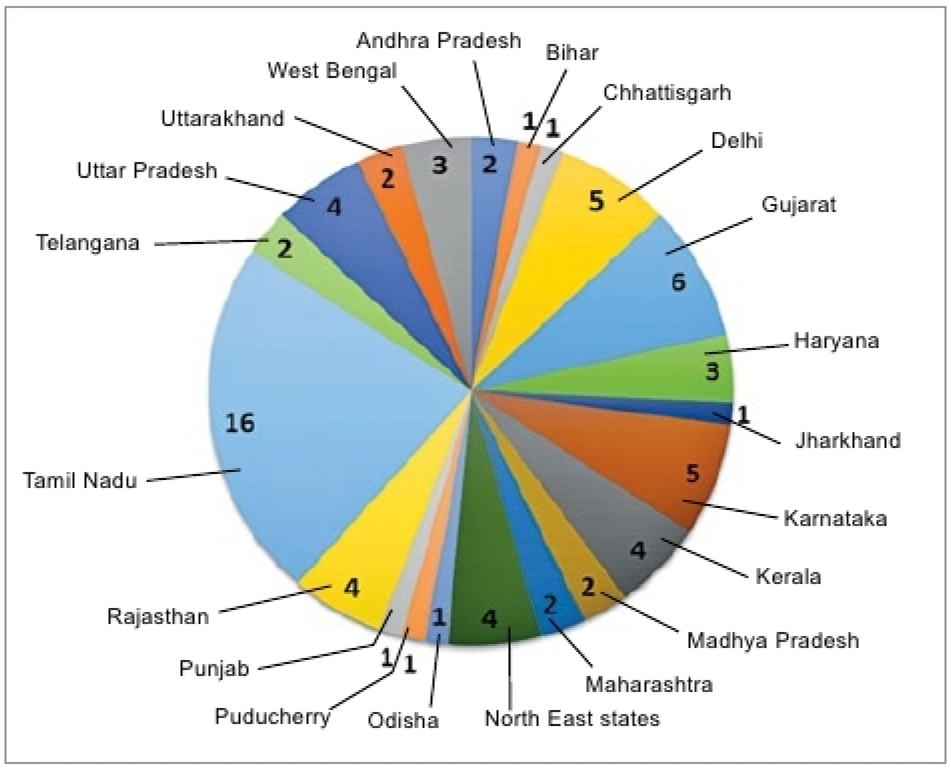

Fig. 2: Number of institutes offering nanotechnology courses in different states in India (43 per cent of the courses are offered in institutions in South India of which 23 per cent are in Tamil Nadu)

12. Kerala Minerals & Metals (KMML): Titanium dioxide nanoparticles

13. Micromaterials (P) Ltd: Nano and micro technologies and materials catalysts; new-generation catalysts are the result of a radically new patented process

14. Mittal Enterprises: Nanofluid

15. Nano Cutting Edge Technology NanoCET: Bio-stabilised nanoparticles

16. Nanomics Technologies: Nanopowder, nanomaterials and suspensions

17. NanoResearch Elements: Nanomaterials

18. Nanoshel: Nanotubes and nanomaterials

19. NanoSniff Technologies: Commercial spin-off from CEN at IIT Bombay; formed to productise technologies developed as part of research work conducted at CEN. This is the first Indian company to successfully commercialise microcantilever and microheater sensor technologies

20. Nanospan: Graphene and instruments

21. NanoXpert Technologies: Nanoparticles

22. Navran Advanced Nanoproducts Development: Polymerised toners

23. Neo-Ecosystems: Metal nanopowders

24. Nilima Nanotechnologies: Nanocoatings

25. NoPo Nanotechnologies: Carbon nanomaterials

26. Platonic Nanotech: High-quality graphene

27. Quantum Corporation: Nanomaterials and nanocomposites as core materials for telecommunications, electronics, drug delivery, conductive films, lighting and energy industries, without the need to change existing processes

28. Reinste Nano Ventures: Nanomaterials

29. Saint-Gobain Glass: High-quality nano-coatings

30. Sisco Research Laboratories (SRL): Chemicals and nanomaterials

31. Smart Nanoz: Nanoparticles

32. Ultrananotech: Nanoparticles and graphene

33. Vecco: Analytical instruments

Future roadmap

The scientific community, policy makers, apex bodies and funding agencies are continuously striving to progress in various areas of nanoscience and nanotechnology, in the fields of water, clean energy, defence, health, medicine, infrastructure, communications and more areas of topical interest. In the last two decades, India has achieved many milestones in the areas of nanoelectronics, nanomaterials and nano-biotechnology, and many are yet to be achieved. It has created many scientists of international repute in the areas of nanoscience and technology, while many young minds are still being formed.

Initiatives like Digital India, Make in India, Startup India, Skill India and other innovative and ambitious programmes like Nanomission, INUP - I and II, and IMPRINT - I and II have given a great thrust to Indian nanotechnology programmes. EFY